ORACLE

How OCI can support FinOps

Operations Advisory Team | Office of CTO

EMEA Technology Cloud Engineering

March 2026

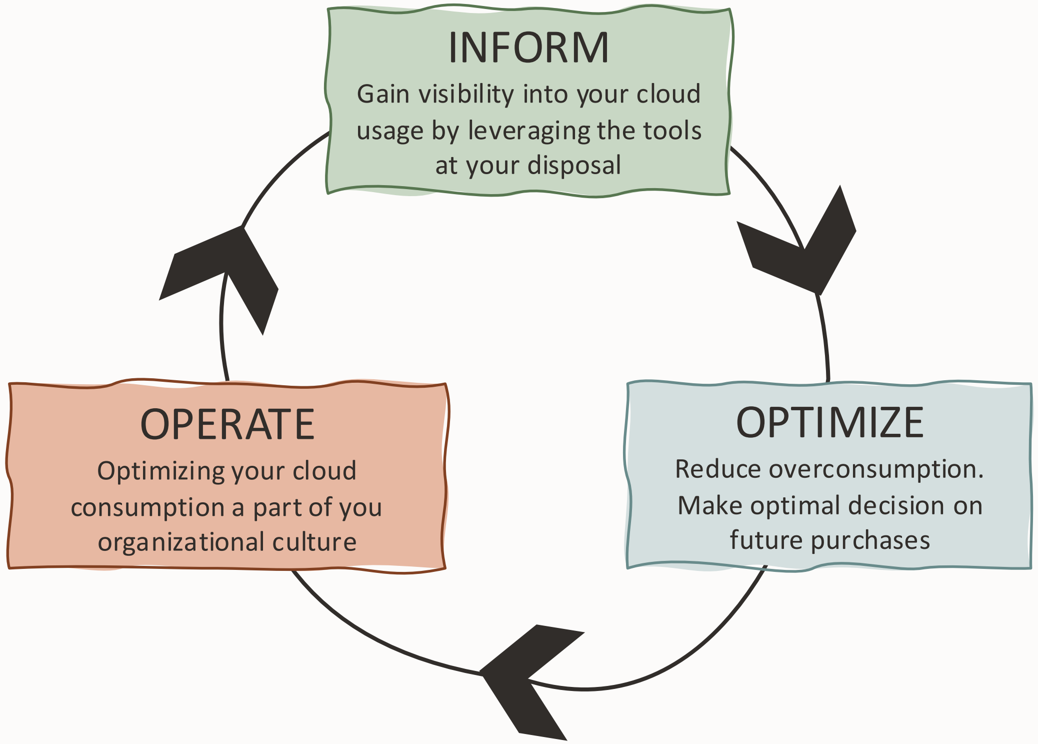
Agenda

Introduction

What is FinOps?

"FinOps is an evolving cloud financial management discipline and cultural practice that enables organizations to get maximum business value by helping engineering, finance, technology and business teams to collaborate on data-driven spending decisions"

FinOps Foundation Technical Advisory Council



FinOps Framework by FinOps Foundation

FinOps Maturity Stages

Crawl

- ✓ Tagging design
- ✓ TCO
- ✓ Billing details by LOB
- ✓ Tracking of cloud spending
- ✓ Monthly trend analysis
- ✓ Forecasting

Walk

- ✓ Cost allocation mapping
- ✓ Tagging in place
- ✓ Dashboards
- ✓ Details available lot LoB leaders

Run

- ✓ Chargeback based on actual consumption
- ✓ Automation of usage monitoring
- ✓ Cloud service catalog

Cost Allocation Method

Direct Costs vs Shared Cost

Shared Cost Allocation

Proportional:
Shared cost allocated based on the direct cost proportion

Preferred by small organizations

Even Split:
Shared costs are allocated evenly across LOB

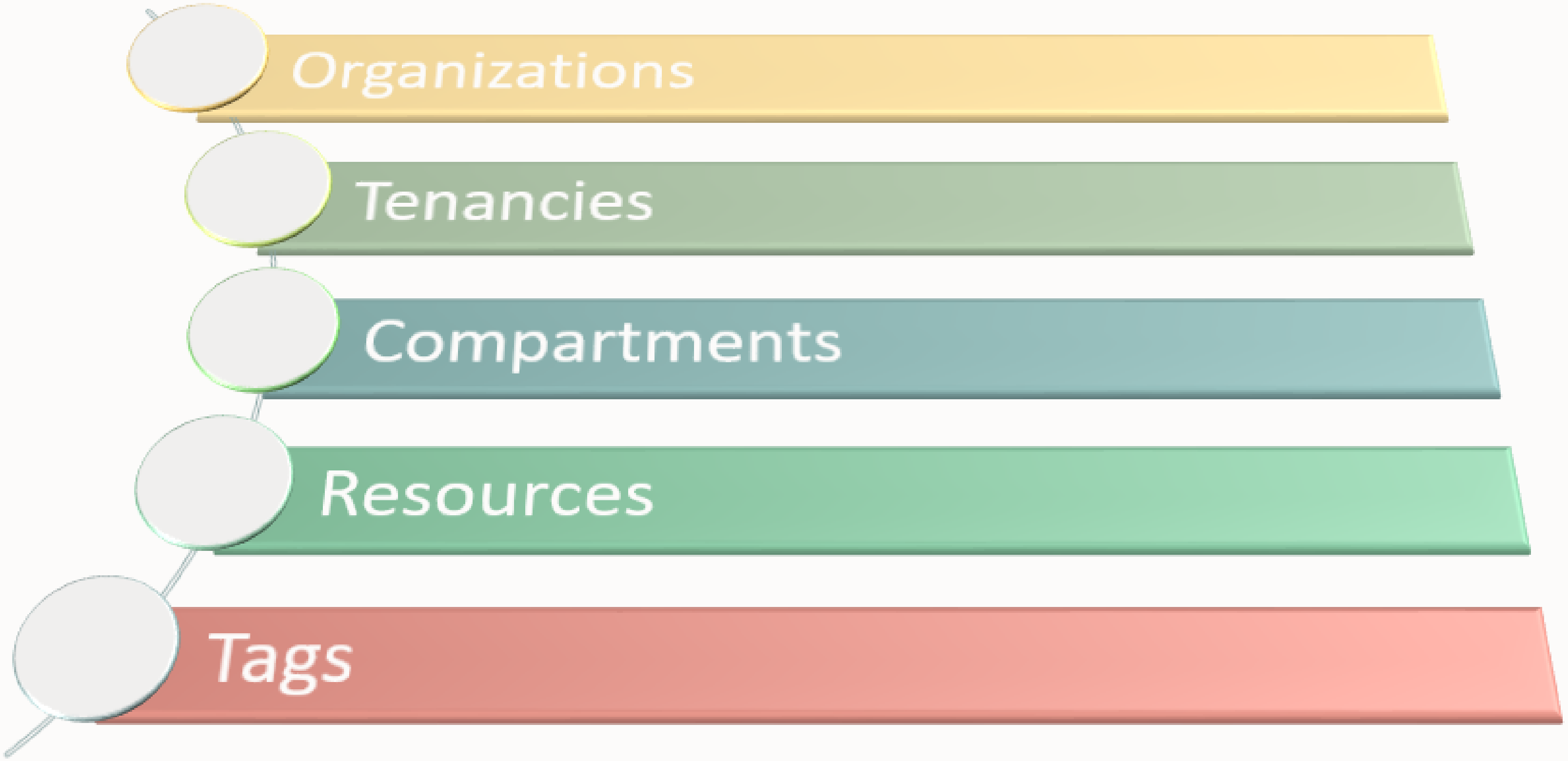
Fixed:
Allocated based on a coefficient

- If shared costs are not allocated effectively and there is no accountability to manage these shared costs
- More mature organizations may need a greater level of sophistication

Common FinOps KPIs

- ✓ Percentage of Costs Associated with Untagged CSP Resources
- ✓ Percentage of Costs Associated with Unallocated CSP Resources
- ✓ Percentage Resource Utilization
- ✓ Percentage Variance of Budgeted vs. Actual CSP Cloud Spend

OCI Constructs to make FinOps easy



Cost reporting Tools are aware of resource hierarchy.

Make a plan that will work for your organization and allow proper cost attribution.

For example, map tenancies with your organizational structure and compartments with scope, TAGs with projects

FinOps Lifecycle

FinOps Lifecycle

Inform

- **Cost Analysis:**
  - Group By/Filter by Comparments/Tags/Services.
  - Export Data (CSV, PDF)
  - Estimate future usage based on current patterns
- **Custom Cost Reports**
- **Budgeting:** Trigger email notification
- **Subscriptions**

Optimize

- **Cloud Advisor**:
  - Identify optimization and cost-saving opportunities -> ‘Fix-it’flow
  - Customize recommendations
- **Budgeting:**
  - Integration with events to automate the creation of Quotas
- **Resource Scheduler**

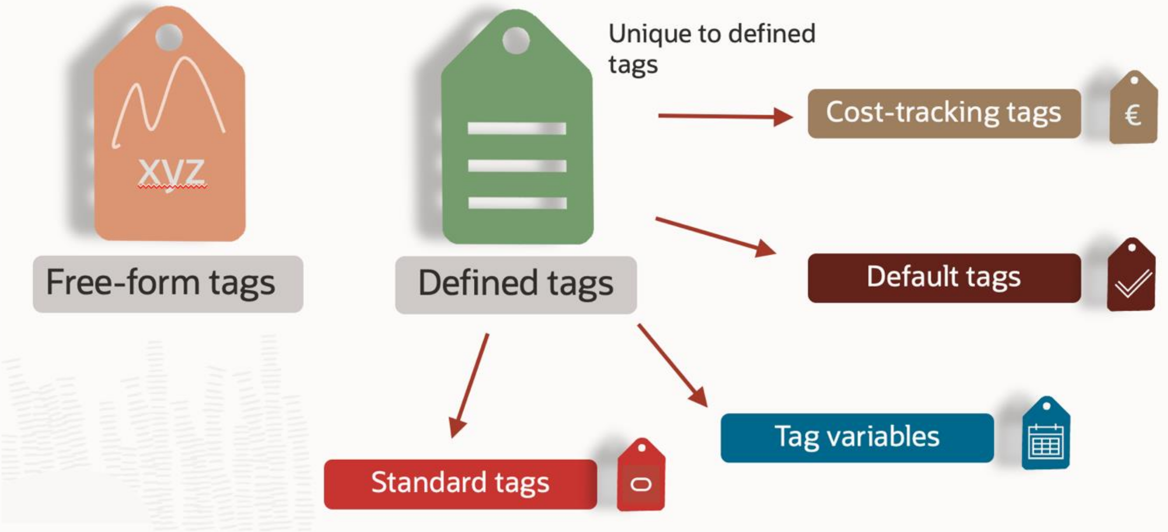
Operate

- Define Shared cost allocation model
- Keep on refining cost control practices
- Multi-cloud cost analysis (**FOCUS**)
- Tenancy management
- Organizational Governance
- **Tags**

FinOps HUB

OCI Tools and Services

Tagging



Cost Tracking Tags

Defined tag

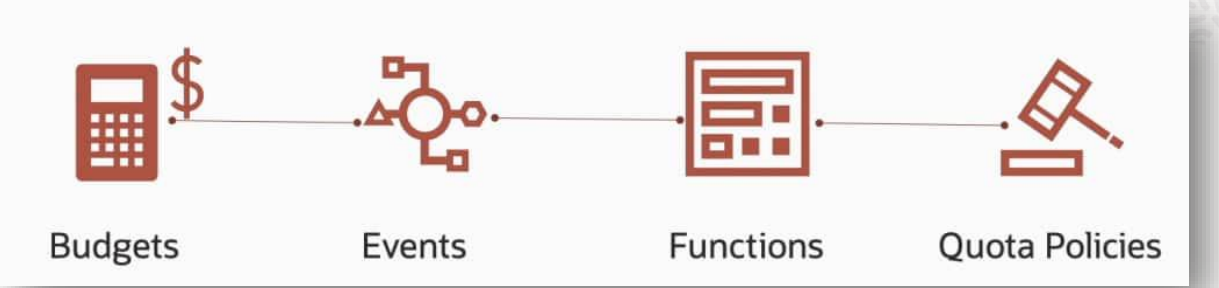
Cost tracking is a feature of defined tags

Budgets

When a tag is designated as a cost-tracking tag, you can set a Budget to track all spending in that cost-tracking tag

https://docs.oracle.com/en-us/iaas/Content/Tagging/home.htm

Budget



- A budget can be used to set soft limits on the Oracle Cloud Infrastructure spending by setting cost-tracking tags or by compartments (including the root compartment) to track all spending in that cost-tracking tag or for that compartment and its children.
- All budgets and spending limits are available from one single place in the Oracle Cloud Infrastructure console.
- All budget alerts are evaluated every 24 hours.
- Budgets help you track your Oracle Cloud Infrastructure (OCI) spending. They monitor costs at a compartment level or cost-tracking tag level. You can set alerts on a budget to receive an email notification based on an actual or forecasted spending threshold. Budget alerts also integrate with the Events service. You can use this integration and the Oracle Notifications service to send messages through PagerDuty, Slack, or SMS.*

www://

https://docs.oracle.com/en-us/iaas/Content/Billing/Concepts/budgetsoverview.htm

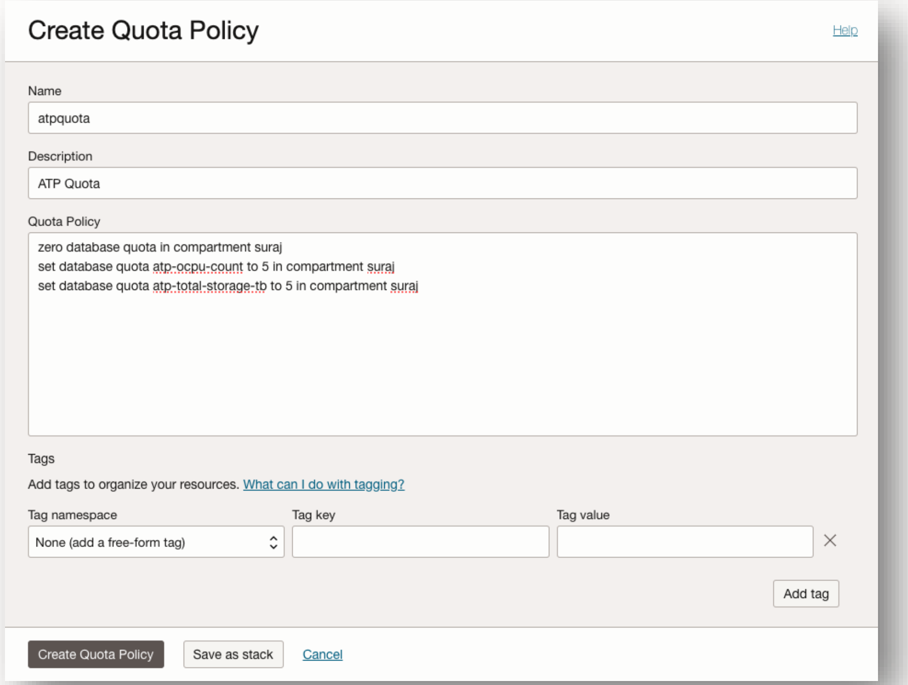
Compartment Quotas

- Compartment quotas are policies that allow administrators to allocate resources with a high level of flexibility by giving tenant and compartment administrators better control over how resources are consumed in OCI, creating a powerful toolset to manage spending within the tenancy.
- Compartment quotas are set by administrators, using policies that allow them to allocate resources with a high level of flexibility.

Set defines the max number of a cloud resource that a compartment can use

Unset reverts quotas to default service limits

Zero denies access to a cloud resource for a compartment



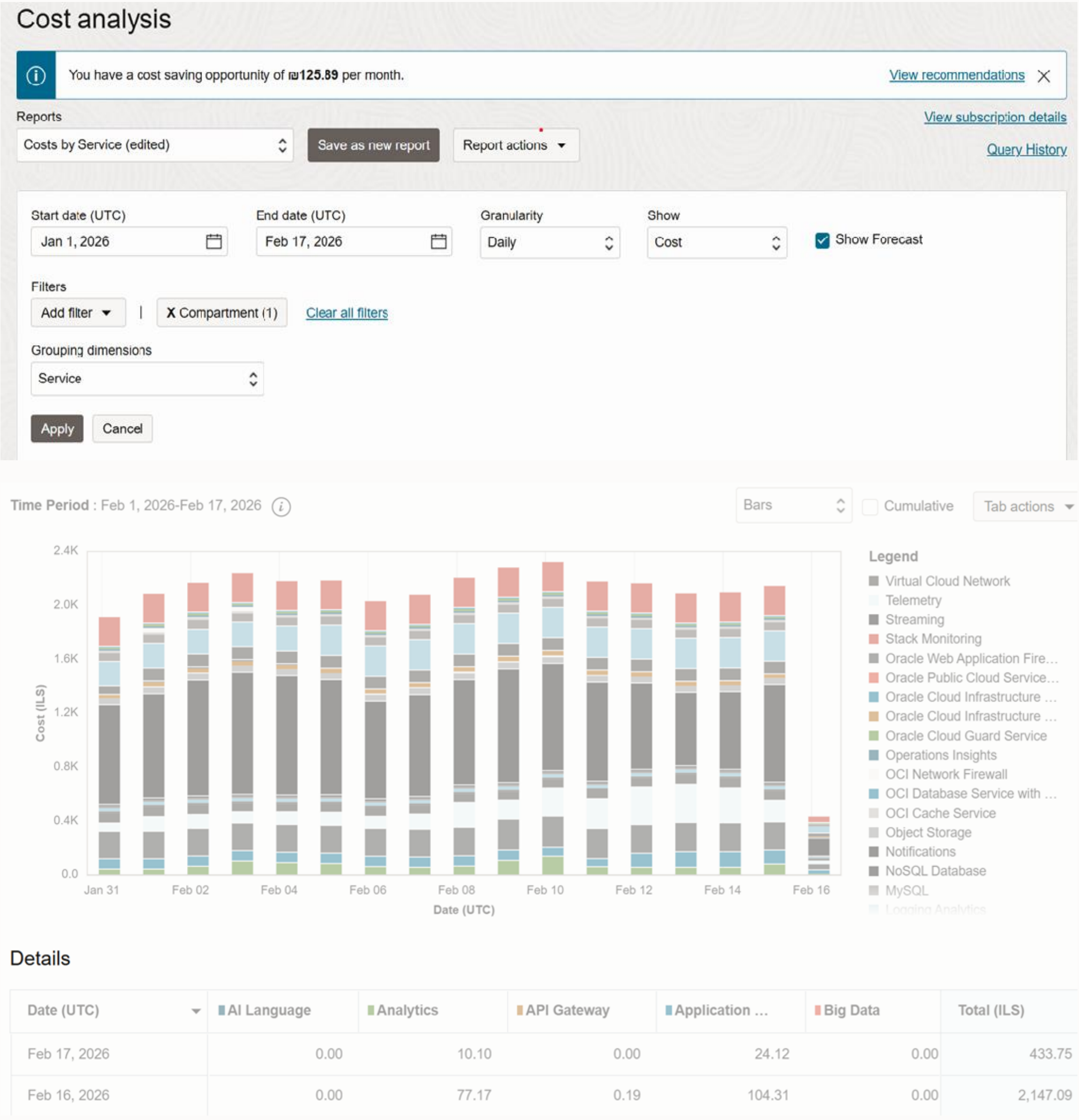
https://docs.oracle.com/en-us/iaas/Content/Quotas/home.htm

OCI Cost Analysis

Menu → Billing & Cost Management →
Cost Management → Cost Analysis

Helps track and optimize your Oracle Cloud Infrastructure spending

- Easy-to-use **visualization tool**
- Filter costs by date, tags and buckets and export it
- Trend lines show how spending habits change
- Forecast: Estimate future consumption and costs



https://docs.oracle.com/en-us/iaas/Content/Billing/Concepts/costanalysisoverview.htm

Cost and Usage Reports

Menu ➔ Billing & Cost Management ➔
Cost Management ➔ Cost and Usage Reports

One or more CSV files (OCI or Focus Report Format) per day with:

- Meta data (OCI: compartment, region, etc.- Focus: Availability Zone)
- Consumption (product/service, cost, unit cost, quantity, etc.)
- Tags (tags, etc.) ➔ Filter costs by tags

➢ Cost and Usage reports can contain corrections from past consumption.
➢ Currently OCI is on Focus version 1.1 and will support 1.3 in future.
➢ Timeline for 1.3 is subject to confirmation from product management.

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O
1	lineItem/referenceNo	lineItem/tenantId	lineItem/intervalUsageStart	lineItem/intervalUsageEnd	product/service	product/compartmentId	product/compartmentName	product/region	product/availabilityDomain	product/resourceId	usage/billedQuantity	usage/billedQuantityOverage	cost/subscriptionId	cost/productSku	product/Description
2	V2.3eYCk8/m	ocid1.tena	2021-09-14T21:00Z	2021-09-14T22:	ANALYTICS_CLOU	ocid1.compartment.oc1..	CloudNativePod	eu-frankfurt-1		ocid1.analyticsinstance	1.000000000000	1.000000000000	7185035	B89631	Oracle Analytics Cloud - Enterprise
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8	V2.3eYCk8/m	ocid1.tena	2021-09-14T21:00Z	2021-09-14T22:	ANALYTICS_CLOU	ocid1.compartment.oc1..	AnalyticsDemoPod	eu-frankfurt-1		ocid1.analyticsinstance	0.150000000000	0.150000000000	7185035	B89637	Oracle Analytics Cloud - Enterprise - BY
9	V2.3eYCk8/m	ocid1.tena	2021-09-14T21:00Z	2021-09-14T22:	ANALYTICS_CLOU	ocid1.compartment.oc1..	AnalyticsDemoPod	eu-frankfurt-1		ocid1.analyticsinstance	2.000000000000	2.000000000000	7185035	B89637	Oracle Analytics Cloud - Enterprise - BY
10	V2.3eYCk8/m	ocid1.tena	2021-09-14T21:00Z	2021-09-14T22:	ANALYTICS_CLOU	ocid1.compartment.oc1..	AAP-Anis.zerelli	eu-frankfurt-1		ocid1.analyticsinstance	1.000000000000	1.000000000000	7185035	B89637	Oracle Analytics Cloud - Enterprise - BY

BI/ reporting dashboard

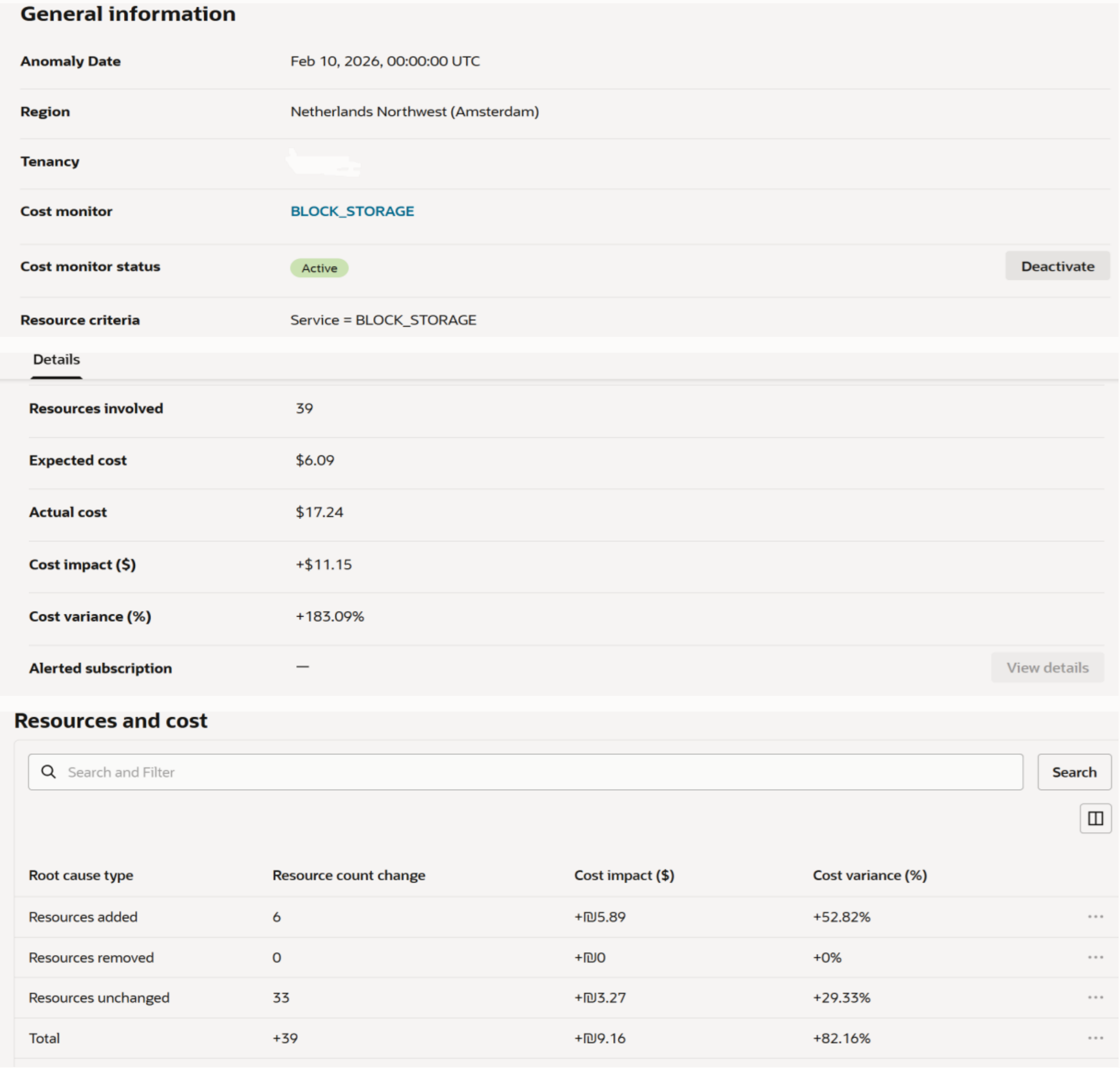
https://docs.oracle.com/en-us/iaas/Content/Billing/Concepts/costusagereportsoverview.htm

Cost Anomaly Report

Menu ➔ *Billing & Cost Management* ➔
Cost Management ➔ ***Cost Anomaly Detection***

Helps to identify abnormal usage patterns across tenancies

- Requires 60 days of historical data before it becomes active
- Oracle defined monitors are available for each region, tenancy and service combination
- Subscribe to an anomaly alert for notification by email whenever an anomaly is detected
- Uses machine learning considering yearly, weekly, and daily seasonality plus holiday effects to forecast daily cost data and automatically detect anomaly events. It learns from user provided feedback improving its accuracy.



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Cloud Advisor

Cost management

- Underutilized compute instances
- Overprovisioned Autonomous Data Warehouse instances
- Unattached block volumes, unattached boot volumes
- Object Storage buckets without lifecycle policy rules

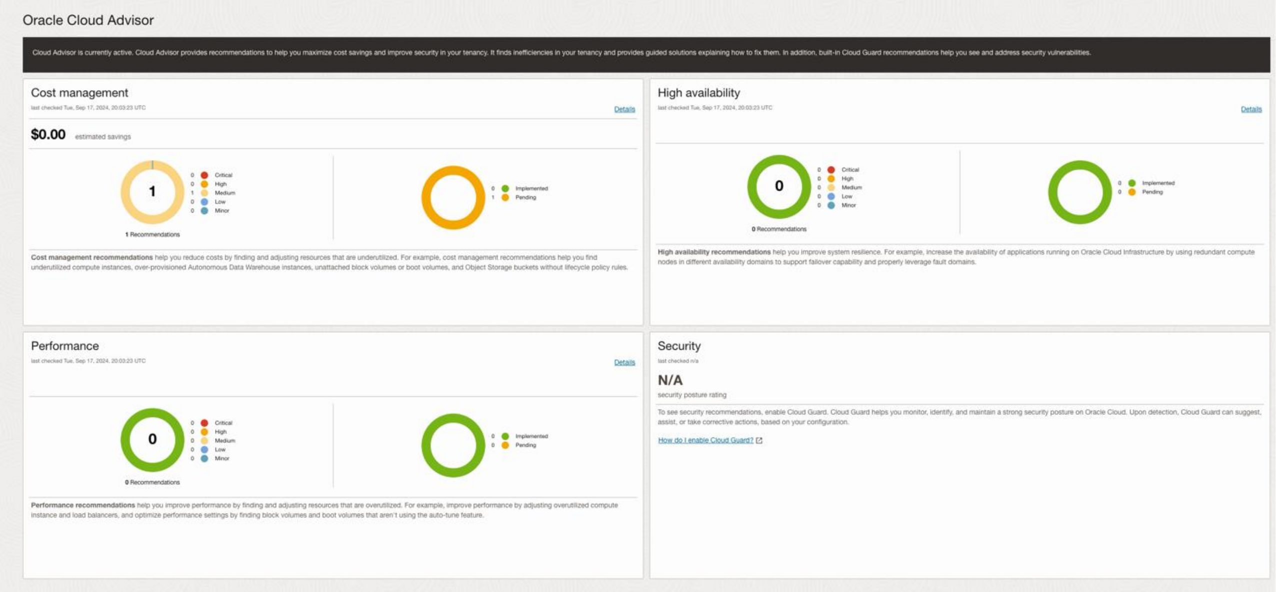
Performance

- Rightsizing overutilized compute instances and load balancers
- Optimize performance settings by finding block volumes and boot volumes that aren't using the autotune feature.

High availability

- Hardware failure best practices to ensure the resilience of your solution and show you how to improve system resilience

Menu ➔ Governance & Administration ➔ ***Cloud Advisor***



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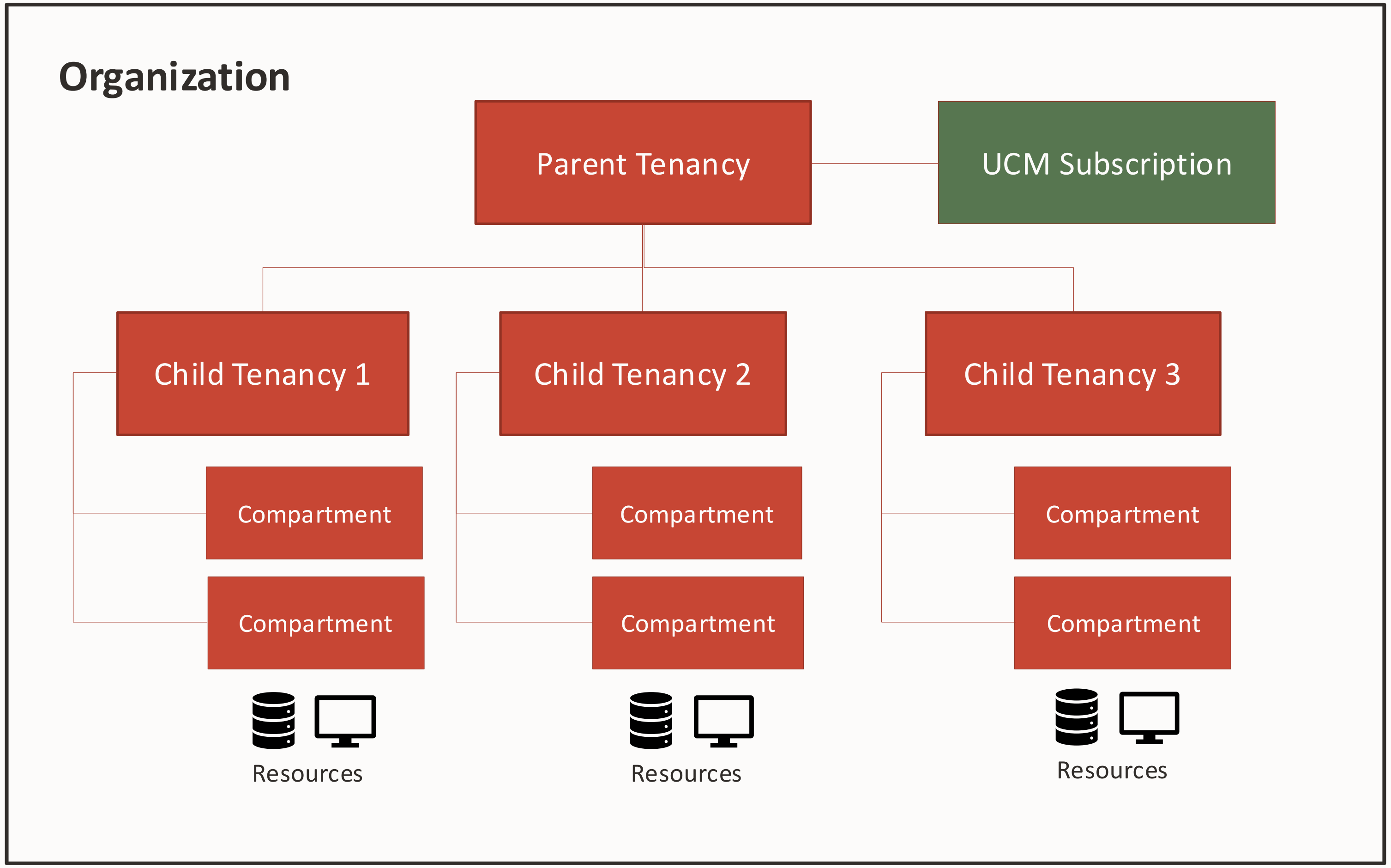
The Organizations model



Organizations enables customers to build a multi-tenancy environment under a single subscription.

Child tenancies are isolated partitions, meaning their one tenancy cannot access another tenancy by default.

The Parent tenancy can centrally manage costs and governance.

- Cost Analysis/Cost Reports
- Budgets on Child Tenancy/ Compartment/ Tag
- Cloud Advisor
- Governance Rules
  - Quotas
  - Regions
  - Tags

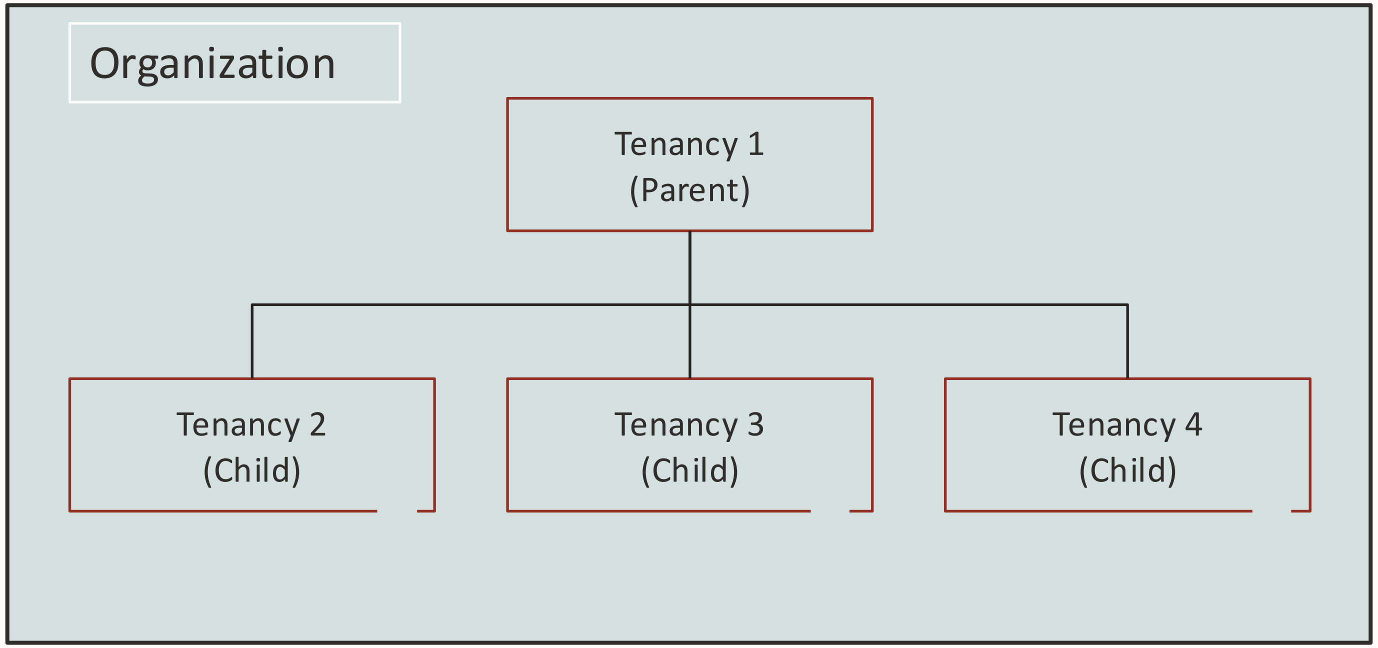
https://docs.oracle.com/en-us/iaas/Content/General/organization/organization_management_overview.htm

Organizations – Planning Considerations

- ➢ The main reason to have multiple tenancies is for strong isolation, to help isolate workloads.
- ➢ Managing multiple tenancies can create extra management overhead:
  - ✓ Ensure that the isolation is worth it.
  - ✓ If you don't require a strong isolation level, consider using compartments to separate workloads.
- ➢ By default, each parent and child tenancy comes with:
  - A distinct set of IAM users (which can be federated to another identity system).
  - A distinct set of IAM policies (permissions).
  - A distinct tenancy administrator.
  - Its own service limits.
  - Isolated Virtual Cloud Networks (VCNs).
  - Separate security and governance settings.

Introducing Organizational Governance

Organizational Governance gives customers assurance that enforcements are created within their org.

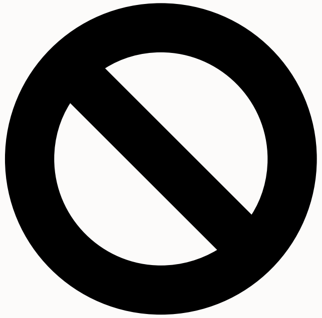


Integrated Services

- Service quotas
- Allowed regions
- Tagging

https://docs.oracle.com/en-us/iaas/Content/General/organization/home.htm

Leverage Existing Governance Controls



Service quotas

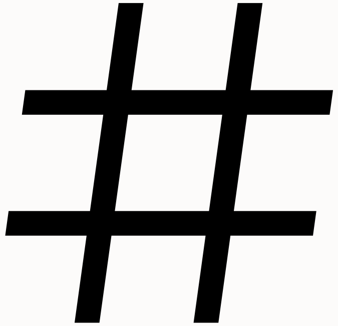
Quotas restrict and limit the usage of services and SKU.

They help prevent excessive resource provisioning by restricting the number of resources (e.g., compute instances) a child tenancy can create.

Allowed regions

Allowed regions set the regions a child tenancy can subscribe to.

This ensures that data resides in only regions which are compliant to IT standards set by the Organization Administrator



Tagging

Tag namespaces and default tags enforce consistent resource tagging across the organization.

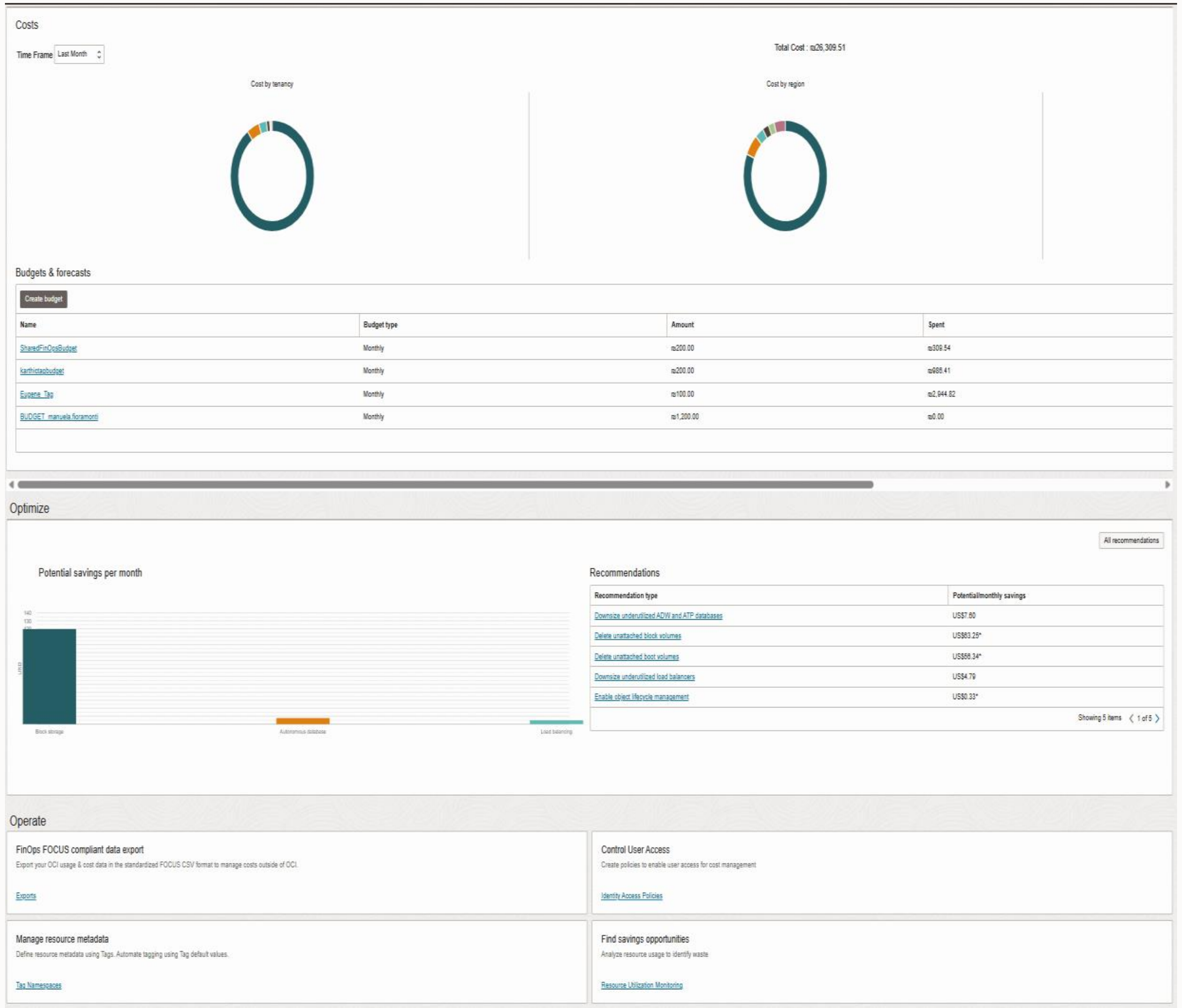
This enables accurate cost allocation, reporting, and financial management.

FinOps Hub

Menu ➔ Billing & Cost Management ➔
Cost Management ➔ Overview

The FinOps Hub page is organized in terms of the following sections:

- Inform
  - Subscriptions
  - Costs
  - Budgets and Forecasts
- Optimize
- Operate



https://docs.oracle.com/en-us/iaas/Content/Billing/Concepts/FinOps.htm

Cost Management Framework

Parent Tenancy

↓

TAG defaults	Resource Quota	Region
TAG Namespace	Budget	Cloud Advisor

↓

Child Tenancy

	Capabilities	OCI services
Organize	Structure cloud resources	Organization, Tenancies, Compartment, Tagging
Report & Analyze	Track cost & usage	Cost Analysis, Cost reports
Governance	Guideline & policy compliance	Quotas, Organization Management, Quotas, Governance rules
Budget & Forecast	Proactive spend management	Budgets
Optimize & Reduce Cost	Rightsize and offset support cost	Cloud advisor, Support rewards

OCI FinOps Tools Matrix

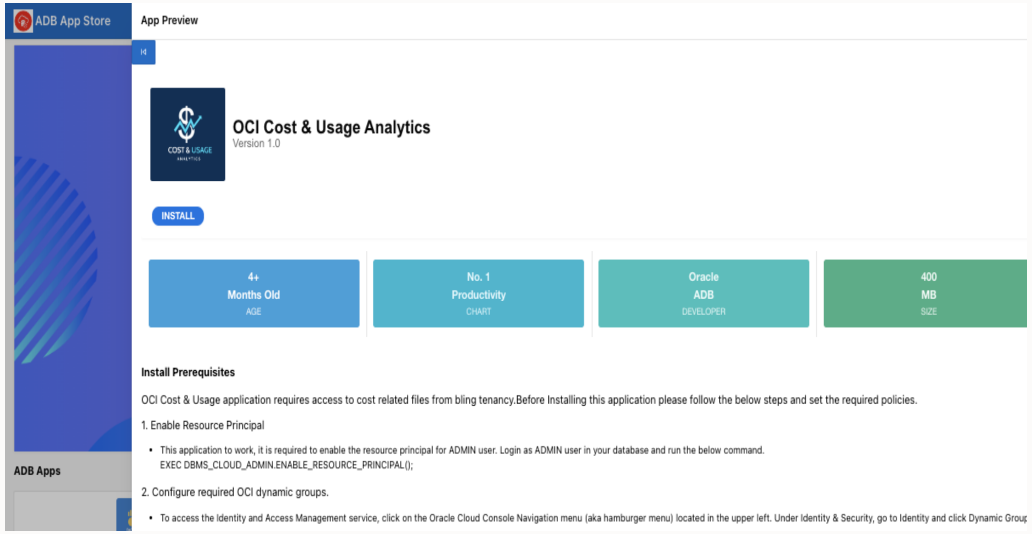
Capability	Tool
Cloud cost planning	OCI Cost Estimator
Billing and reporting	Cost Analysis Cost and usage reports
OCI **FinOps Hub** page	Using OCI FinOps Hub
Invoicing	Invoices Payment history Subscriptions & Billing schedule
Forecasting	Forecasting in Cost Analysis
Tagging	Tags
Alerts and notifications	Budget alerts
Controls	Quotas Enforcing budgets using functions and quotas
Recommendations	Cloud Advisor
Other Tools	Oracle Analytics Cloud (OAC) OCI Cost & Usage Application

Basic Org & Child Tenancy Questions

FAQ	RESPONSE
What is a child tenancy?	A child tenancy is an isolated partition of OCI. The parent tenancy cannot access the child tenancy except to view cost management information.
Who can add a child tenancy?	A parent must have a UCM subscription and the right limits. Limits can be requested through the console.
Who can join as a child tenancy?	A stand alone tenancy with either a PAYG, UCM Commit, or Funded Allocation subscription can join an Org.
Who can view costs?	The parent tenancy can view costs across all tenancies. The child tenancy can only view their own costs.
What services can the parent use across the Organization?	Cost Analysis, Cost Reports, Budgets, Cloud Advisor, Subscription Mapping, Quotas, Region Subscription, Tags
I can't view my invitation?	Check to make sure the child tenancy is in its home region.
I can't create a tenancy or send an invitation.	First check to make sure you have a UCM subscription. If you do, its likely it was upgraded without a limits increase, so open an SR for a limits increase.
I am using a PAYG sub, but I want to use Orgs	This is evaluated on a case by case basis. Open a ticket and the CAM team will review.
Can I delete a child tenancy?	Yes, you can delete from the child tenancy. Go to Tenancy Details -> Delete Tenancy. Deleting a tenancy puts it into soft termination for 30 days. Then the resources will be terminated.
Can I remove (unlink) a child tenancy from an Org?	Yes, as the parent you can remove the child tenancy if it was linked to the org. Go to subscription mapping and map the child back to its original sub. Then go to tenancy and remove tenancy.
What subscription is my tenancy using?	You can go to the customer's subscription mapping page to view the assignment.

Next Steps

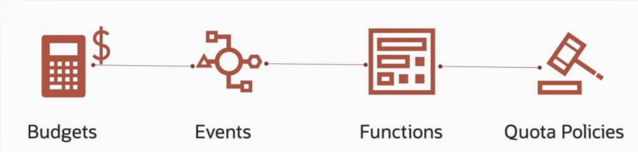
FinOps Approach: Tackling Inform and Optimize Phase



- Identify 1 or 2 main concerns related to financial management for one customer and automation needs
- Advise on Tagging/Budgeting

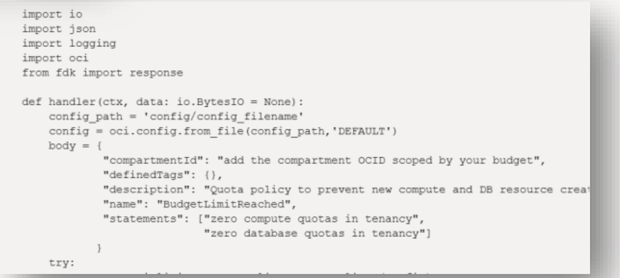
Support Prototyping a detailed cost analysis process:

- A cost report based on identified needs
- Download and Import for analysis via ADB
- Function for quota creation



Advise on additional automation possibilities

Event type: Create Triggered Alert



```
import io
import json
import logging
import oci
from fdk import response

def handler(ctx, data: io.BytesIO = None):
    config_path = 'config/config_filename'
    config = oci.config.from_file(config_path,'DEFAULT')
    body = {
            "compartmentId": "add the compartment OCID scoped by your budget",
            "definedTags": {},
            "description": "Quota policy to prevent new compute and DB resource creat
            "name": "BudgetLimitReached",
            "statements": ["zero compute quotas in tenancy",
                           "zero database quotas in tenancy"]
           }
    try:
```

Next Steps

- Management and chargeback
- Technical Optimization
- Are there any shared costs that need to be allocated?
- Automation requirements?
- Workshop Tagging for chargeback and cost management
- Forecast usage and spend

Stay tuned

[Operations Advisory GitHub Repository]

Thank you

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